~~Title~~Stubbornness and Social Power in Opinion Dynamics: the French–Harary–DeGroot and Friedkin–Johnsen Models

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Abstract

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1 Introduction

How do the opinions of a group evolve when its members listen to each other, and what changes when some of them resist? Models of opinion dynamics describe a social network as a weighted digraph, whose nodes are the agents and whose edges carry the influence of one agent on another, and update the opinions by repeated averaging [5]. This report compares two discrete-time models of this kind, following the tutorial by Proskurnikov and Tempo [5], the course slides [2, 3, 1] and, as a support, the book by Bullo [4].

- In the *French–Harary–DeGroot* (FDG) model every agent replaces its opinion with a weighted average of the opinions it listens to, its own included. The outcome depends only on the network: consensus, disagreement or no convergence, and, under consensus, the social power of each agent, i.e. the weight of its initial opinion in the common final opinion.
- In the *Friedkin–Johnsen* (FJ) model every agent also keeps an attachment to a fixed prejudice. The prejudices act as a constant input and, when every agent is reached by a prejudice, the final opinions are convex combinations of them.

The thread of the report is stubbornness: an agent that never changes its opinion, and the graded mechanisms of which it is the extreme case, a self-weight in the FDG model and a prejudice in the FJ model. We ask how these mechanisms change three things: whether the opinions converge, what they converge to, and how much each agent weighs in the result, measured by the social power in the FDG model and by the influence centrality in the FJ model. The two models are compared on the same small networks, with agent 1 as the agent that resists, so that every difference comes from the model and not from the data.

The tutorial [5] also treats two continuous-time models, which play the same roles in continuous time. Abelson's model, $\dot{x}_i = \sum_j a_{ij}(x_j - x_i)$, is the counterpart of the FDG model; Taylor's model, $\dot{x}_i = \sum_j a_{ij}(x_j - x_i) + \gamma_i(u_i - x_i)$, is the counterpart of the FJ model: an agent with $\gamma_i > 0$ is prejudiced, and the notion of P-dependent agent, central in Section 3, first appears there [5]. They are mentioned only as an overview and are not studied in this report.

Section 1.1 recalls the notions on graphs and non-negative matrices used throughout. Section 2 introduces the FDG model and studies stubbornness and social power in it; Section 3 does the same for the FJ model, from its convergence and the augmented graph to the influence centrality; Section 4 compares the two models.

1.1 Preliminaries

This subsection collects the notions on graphs and non-negative matrices used for both models, following the course slides [2, 3, 1].

1.1.1 Graphs

A *weighted digraph* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ has node set $V = \{1, 2, \dots, n\}$, edge set $E \subseteq V \times V$ and a weight $a_{ij} > 0$ for each edge $(i, j) \in E$; an edge (i, i) is a *self-loop*. Throughout the report the nodes are agents, and an edge (i, j) means that agent i is influenced by agent j. This is the convention of the course slides [1]; the tutorial by Proskurnikov and Tempo draws the edges in the opposite direction [5].

Definition 1.1 (Weighted adjacency matrix [2])**.** Given a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ of order n the (weighted) adjacency matrix of G is the $n \times n$ matrix A in which:

(i) for each edge $(i, j) \in E$ the (i, j) entry of A is equal to the edge weight a_{ij},

(ii) all other entries of A are equal to zero.

Definition 1.2 (Connectivity of digraphs [2])**.** A directed graph $G = (V, E)$ is

(i) *strongly connected* if there is a directed walk from any node to any other node;

(ii) *weakly connected* if the undirected version of the graph is connected.